

■ Two-Year Integrated Curriculum in Physics, History & Philosophy

Module	Core Physics Topics	Primary Sources	Technical Textbooks	Historical / Contextual Sources	Philosophy / Conceptual Focus
I. Ancient & Classical Foundations (Weeks 1–8)	Kinematics, motion, experiment, scientific method	Aristotle – <i>Physics</i> ; Archimedes – <i>On Floating Bodies</i> ; Galileo – <i>Two New Sciences</i> ; Copernicus – <i>On the Revolutions</i>	Halliday, Resnick & Walker – <i>Fundamentals of Physics</i> (Chs. 1–5); Kleppner & Kolenkow – <i>An Introduction to Mechanics</i> (Chs. 1–3)	Grant – <i>A History of Natural Philosophy</i> ; Dijksterhuis – <i>The Mechanization of the World Picture</i> ; Westfall – <i>The Construction of Modern Science</i> ; Koyré – <i>From the Closed World to the Infinite Universe</i>	Origins of scientific reasoning; causation; empiricism; mathematization of nature
II. The Newtonian Revolution (Weeks 9–16)	Laws of motion, gravitation, optics, conservation	Newton – <i>Principia Mathematica</i> (Books I–III); Newton – <i>Opticks</i> ; Leibniz–Clarke Correspondence; Kant – <i>Metaphysical Foundations of Natural Science</i>	Goldstein – <i>Classical Mechanics</i> (Chs. 1–5); Feynman – <i>Lectures on Physics</i> Vol. I	Cohen – <i>Introduction to Newton's Principia</i> ; Westfall – <i>Never at Rest</i> ; Hall – <i>Philosophers at War</i> ; Friedman – <i>Kant and the Exact Sciences</i>	Laws of nature; causality and determinism; a priori foundations; the mechanistic worldview
III. Fields and Energy (Weeks 17–24)	Lagrangian/Hamiltonian mechanics, electromagnetism, thermodynamics	Lagrange – <i>Analytical Mechanics</i> ; Hamilton – “On a General Method...”; Faraday – <i>Experimental Researches in Electricity</i> ; Maxwell – <i>Treatise on Electricity and Magnetism</i> ; Helmholtz – “On the Conservation of Force”; Clausius – “On	Goldstein – <i>Classical Mechanics</i> (Chs. 6–10); Landau & Lifshitz – <i>Mechanics</i> ; Griffiths – <i>Introduction to Electrodynamics</i> ; Schroeder – <i>Thermal Physics</i>	Buchwald – <i>From Maxwell to Microphysics</i> ; Hunt – <i>The Maxwellians</i> ; Jungnickel & McCormmach – <i>Intellectual Mastery of Nature</i> ; Brush – <i>The Kind of Motion We Call Heat</i>	Field ontology; energy conservation; unification of forces; the conceptual meaning of “law”

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		the Moving Force of Heat”			
IV. Relativity & Space-Time Geometry (Weeks 25–32)	Special and general relativity, spacetime structure, cosmology	Einstein – 1905 & 1916 papers; Minkowski – “Space and Time”; de Sitter – “On Einstein’s Theory of Gravitation”	Taylor & Wheeler – <i>Spacetime Physics</i> ; Schutz – <i>A First Course in General Relativity</i> ; Carroll – <i>Spacetime and Geometry</i>	Holton – <i>Thematic Origins of Scientific Thought</i> ; Stachel – <i>Einstein from “B” to “Z”</i> ; Pais – <i>Subtle Is the Lord</i> ; Kragh – <i>Cosmology and Controversy</i>	Geometry and reality; conventionalism; relativity of simultaneity; the epistemology of space-time
V. Quantum Revolution (Weeks 33–44)	Quantum mechanics, uncertainty, measurement, early quantum field theory	Planck (1900); Einstein (1905); Bohr (1913); Schrödinger (1926); Heisenberg (1925); Dirac (1930); Einstein–Podolsky–Rosen (1935); Bohr reply	Griffiths – <i>Introduction to Quantum Mechanics</i> ; Sakurai – <i>Modern Quantum Mechanics</i> ; Dirac – <i>Principles of Quantum Mechanics</i> ; Peskin & Schroeder – <i>An Introduction to Quantum Field Theory</i>	Jammer – <i>Conceptual Development of Quantum Mechanics</i> ; Cassidy – <i>Uncertainty</i> ; Mehra & Rechenberg – <i>Historical Development of Quantum Theory</i> ; Schweber – <i>QED and the Men Who Made It</i>	Probability and indeterminacy; realism vs instrumentalism; complementarity; paradigm shifts (Popper, Kuhn, Feyerabend)
VI. Modern Physics & Cosmology (Weeks 45–52)	Statistical mechanics, particle physics, cosmology, metaphysics of science	Boltzmann – <i>Lectures on Gas Theory</i> ; Feynman – <i>QED</i> ; Hawking – <i>A Brief History of Time</i>	Schroeder – <i>Thermal Physics</i> ; Griffiths – <i>Introduction to Elementary Particles</i> ; Ryden – <i>Introduction to Cosmology</i>	Brush – <i>Kind of Motion We Call Heat</i> Vol. II; Kragh – <i>Conceptions of Cosmos</i> ; Kaiser – <i>Drawing Theories Apart</i> ; Close – <i>The Infinity Puzzle</i>	Reductionism and emergence; realism vs anti-realism; unification of laws; cosmological meaning

Capstone (Weeks 51–52)

Goal	Output Examples
Integrate technical, historical, and philosophical understanding in one	• Research essay (e.g., “Kant and Newton on Necessity in Nature”) • Computational simulation (orbital mechanics, field visualizations, or Schrödinger solver) • Comparative study (e.g.,

Goal	Output Examples
substantial work	“Fields, Geometry, and Quantum Structure”)*

Excellent — a **study rhythm** will keep this interdisciplinary plan balanced between *technical mastery* and *conceptual depth*. Below is a flexible weekly rhythm designed for **10–15 hours per week** (for the 2-year pace). You can double it (20–30 hrs/week) to compress into 1 year.

It balances **problem-solving**, **reading**, and **reflection**, while integrating philosophy and history *without sacrificing mathematical rigor*.

Suggested Weekly Study Rhythm

(Example: 10–15 hours per week; adapt by pace or interest)

Day	Focus	Activities & Goals	Approx. Time
Monday	Technical Foundations	Read new section of physics textbook (e.g. Goldstein, Griffiths, or Schroeder). Take detailed notes on derivations and examples.	2 hrs
Tuesday	Problem Solving I	Work through textbook problems (odd/even sets). Focus on conceptual + computational understanding.	2 hrs
Wednesday	Primary Source Reading	Read historical/primary text (e.g. Galileo, Newton, Einstein). Annotate key arguments and equations.	1.5–2 hrs
Thursday	Philosophical Reflection	Read assigned philosophy/theory text (e.g. Kant, Duhem, Popper, Kuhn). Write a 1–2 paragraph summary or reflection.	1.5 hrs
Friday	Problem Solving II	Continue physics problem sets or simulations; link equations to primary readings.	2 hrs
Saturday	Historical Context & Integration	Read secondary historical material (e.g. Westfall, Kragh). Write short connections between science, context, and philosophy.	2 hrs
Sunday	Synthesis & Review	Review notes; summarize week’s technical, conceptual, and historical insights. Optional: reflective journal or study group discussion.	1–2 hrs

Weekly Flow Summary

- **≈6 hrs/week** → Physics theory + problem-solving
- **≈3 hrs/week** → Historical readings + contextual study
- **≈3 hrs/week** → Philosophy + synthesis/reflection

Goal per week:

- Master one major technical concept (e.g. conservation laws, field equations, Schrödinger equation)
- Understand its **historical genesis** and **philosophical implications**

- Produce one piece of written reflection (half-page–2 pages)

Monthly Milestones

Each month ($\approx$ 4 weeks):

- **Technical checkpoint:** complete 1–2 textbook chapters + assigned problems
- **Primary source:** finish one historical work or substantial excerpt
- **Philosophy/history essay:** 1–2 pages connecting physics + philosophy
- **Optional computational lab:** Python/Matlab simulation or visualization project

Example Daily Micro-Rhythm (for focused days)

Phase	Time	Goal
Focus Study Block	90 min	Read technical material actively; derive formulas by hand
Short Break	10–15 min	Walk/stretch (no screens)
Problem Practice	60 min	Apply the concepts to end-of-chapter problems
Conceptual Reflection	30 min	Write short notes: “What changed in our understanding of nature here?”

Optional Add-ons (advanced or flexible)

- **Weekly “colloquium hour”:** pick one short modern article or video lecture (e.g. Sabine Hossenfelder, Sean Carroll, Rovelli) to connect modern research with classical ideas.
- **Monthly essay:** integrate historical + technical content (e.g. “How Maxwell unified electricity and magnetism”)
- **Quarterly project:** computational or conceptual synthesis (simulate or model a physical system; write a historical-philosophical paper).

Excellent — here’s **Part II: the Complete Bibliography**, organized by category and era/theme, covering **Primary Sources**, **Historical / Secondary Contextual Works**, **Philosophy of Science**, and **Technical Physics Textbooks**.

It’s formatted in **Chicago style** for easy reference or citation. You can treat this as your foundational library for the full 1–2 year program.

Comprehensive Bibliography for the Integrated Physics–History–Philosophy Curriculum

I. Primary Sources in Physics and Natural Philosophy

Ancient & Classical:

- Aristotle. *Physics*. Translated by R. P. Hardie and R. K. Gaye. Oxford: Clarendon Press, 1995.
- Archimedes. *On Floating Bodies*. In *The Works of Archimedes*, edited by T. L. Heath. Cambridge: Cambridge University Press, 1897.
- Ptolemy. *Almagest*. Translated by G. J. Toomer. Princeton: Princeton University Press, 1998.

Scientific Revolution:

- Copernicus, Nicolaus. *On the Revolutions of the Heavenly Spheres*. Translated by Edward Rosen. Baltimore: Johns Hopkins University Press, 1992.
- Galileo Galilei. *Two New Sciences*. Translated by Stillman Drake. Toronto: University of Toronto Press, 1974.
- Descartes, René. *Principles of Philosophy*. Translated by Valentine Rodger Miller and Reese P. Miller. Dordrecht: Reidel, 1983.
- Kepler, Johannes. *Astronomia Nova*. Translated by William H. Donahue. Cambridge: Cambridge University Press, 1992.

Newtonian & Enlightenment:

- Newton, Isaac. *Philosophiae Naturalis Principia Mathematica*. Translated by I. Bernard Cohen and Anne Whitman. Berkeley: University of California Press, 1999.
- Newton, Isaac. *Opticks*. New York: Dover, 1952.
- Leibniz, Gottfried Wilhelm, and Samuel Clarke. *The Leibniz–Clarke Correspondence*. Edited by H. G. Alexander. Manchester: Manchester University Press, 1956.
- Kant, Immanuel. *Metaphysical Foundations of Natural Science*. Translated by Michael Friedman. Cambridge: Cambridge University Press, 2004.

19th Century:

- Lagrange, Joseph-Louis. *Analytical Mechanics*. Translated by Auguste Boissonnade. Dordrecht: Springer, 1997.
- Hamilton, William Rowan. “On a General Method in Dynamics.” *Philosophical Transactions of the Royal Society of London* 124 (1834): 247–308.
- Faraday, Michael. *Experimental Researches in Electricity*. London: Richard Taylor and William Francis, 1839.
- Maxwell, James Clerk. *A Treatise on Electricity and Magnetism*. Oxford: Clarendon Press, 1873.
- Helmholtz, Hermann von. “On the Conservation of Force.” In *Science and Culture*, edited by David Cahan. Chicago: University of Chicago Press, 1995.

- Clausius, Rudolf. “On the Moving Force of Heat.” In *The Mechanical Theory of Heat*. London: John van Voorst, 1867.

Modern Physics:

- Einstein, Albert. *Relativity: The Special and the General Theory*. Translated by Robert W. Lawson. New York: Crown, 1961.
- Einstein, Albert. *The Collected Papers of Albert Einstein*, Vols. 2–6. Princeton: Princeton University Press, 1987–1998.
- Minkowski, Hermann. “Space and Time.” *Annalen der Physik* 47 (1909): 927–38.
- Planck, Max. “On the Theory of the Energy Distribution Law of the Normal Spectrum.” *Verhandlungen der Deutschen Physikalischen Gesellschaft* 2 (1900): 237–45.
- Bohr, Niels. *Atomic Physics and Human Knowledge*. New York: Wiley, 1958.
- Heisenberg, Werner. *Physics and Philosophy: The Revolution in Modern Science*. New York: Harper, 1958.
- Schrödinger, Erwin. “An Undulatory Theory of the Mechanics of Atoms and Molecules.” *Physical Review* 28 (1926): 1049–70.
- Dirac, Paul A. M. *The Principles of Quantum Mechanics*. 4th ed. Oxford: Clarendon Press, 1958.
- Einstein, Albert, Boris Podolsky, and Nathan Rosen. “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?” *Physical Review* 47 (1935): 777–80.
- Boltzmann, Ludwig. *Lectures on Gas Theory*. Translated by Stephen Brush. Berkeley: University of California Press, 1964.
- Feynman, Richard. *QED: The Strange Theory of Light and Matter*. Princeton: Princeton University Press, 1985.
- Hawking, Stephen. *A Brief History of Time*. New York: Bantam Books, 1988.

II. Historical and Contextual Secondary Sources

General & Foundational:

- Grant, Edward. *A History of Natural Philosophy: From the Ancient World to the Nineteenth Century*. Cambridge: Cambridge University Press, 2007.
- Dijksterhuis, E. J. *The Mechanization of the World Picture*. Oxford: Clarendon Press, 1961.
- Westfall, Richard. *The Construction of Modern Science*. Cambridge: Cambridge University Press, 1977.

Scientific Revolution & Enlightenment:

- Koyré, Alexandre. *From the Closed World to the Infinite Universe*. Baltimore: Johns Hopkins University Press, 1957.

- Westfall, Richard. *Never at Rest: A Biography of Isaac Newton*. Cambridge: Cambridge University Press, 1980.
- Cohen, I. Bernard. *Introduction to Newton's Principia*. Cambridge: Cambridge University Press, 1971.
- Hall, A. Rupert. *Philosophers at War: The Quarrel between Newton and Leibniz*. Cambridge: Cambridge University Press, 1980.
- Friedman, Michael. *Kant and the Exact Sciences*. Cambridge: Harvard University Press, 1992.

19th Century:

- Buchwald, Jed Z. *From Maxwell to Microphysics: Aspects of Electromagnetic Theory in the Last Quarter of the Nineteenth Century*. Chicago: University of Chicago Press, 1985.
- Hunt, Bruce J. *The Maxwellians*. Ithaca: Cornell University Press, 1991.
- Jungnickel, Christa, and Russell McCormmach. *Intellectual Mastery of Nature: Theoretical Physics from Ohm to Einstein*. Chicago: University of Chicago Press, 1986.
- Brush, Stephen G. *The Kind of Motion We Call Heat*. Amsterdam: North-Holland, 1976.

Relativity & Early 20th Century:

- Holton, Gerald. *Thematic Origins of Scientific Thought: Kepler to Einstein*. Cambridge: Harvard University Press, 1973.
- Stachel, John, ed. *Einstein from "B" to "Z"*. Boston: Birkhäuser, 2002.
- Pais, Abraham. *Subtle Is the Lord: The Science and the Life of Albert Einstein*. Oxford: Oxford University Press, 1982.
- Kragh, Helge. *Cosmology and Controversy: The Historical Development of Two Theories of the Universe*. Princeton: Princeton University Press, 1996.

Quantum Revolution:

- Jammer, Max. *The Conceptual Development of Quantum Mechanics*. New York: McGraw-Hill, 1966.
- Cassidy, David C. *Uncertainty: The Life and Science of Werner Heisenberg*. New York: Freeman, 1992.
- Mehra, Jagdish, and Helmut Rechenberg. *The Historical Development of Quantum Theory*. 6 vols. New York: Springer, 1982–2000.
- Schweber, Silvan S. *QED and the Men Who Made It: Dyson, Feynman, Schwinger, and Tomonaga*. Princeton: Princeton University Press, 1994.

Modern Physics & Cosmology:

- Kragh, Helge. *Conceptions of Cosmos: From Myths to the Accelerating Universe*. Oxford: Oxford University Press, 2007.

- Kaiser, David. *Drawing Theories Apart: The Dispersion of Feynman Diagrams in Postwar Physics*. Chicago: University of Chicago Press, 2005.
 - Close, Frank. *The Infinity Puzzle: Quantum Field Theory and the Hunt for an Orderly Universe*. New York: Basic Books, 2011.
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III. Philosophy of Science and Conceptual Sources

Foundations and Classical Philosophy:

- Aristotle. *Posterior Analytics*. Oxford: Clarendon Press, 1991.
- Descartes, René. *Discourse on Method and Meditations on First Philosophy*. Indianapolis: Hackett, 1998.
- Kant, Immanuel. *Critique of Pure Reason*. Translated by Paul Guyer and Allen Wood. Cambridge: Cambridge University Press, 1998.

19th to Early 20th Century:

- Duhem, Pierre. *The Aim and Structure of Physical Theory*. Princeton: Princeton University Press, 1954.
- Mach, Ernst. *The Science of Mechanics*. La Salle, IL: Open Court, 1960.
- Cassirer, Ernst. *Substance and Function and Einstein's Theory of Relativity*. Chicago: Open Court, 1923.

20th Century to Contemporary:

- Popper, Karl. *The Logic of Scientific Discovery*. London: Routledge, 1959.
 - Kuhn, Thomas S. *The Structure of Scientific Revolutions*. 2nd ed. Chicago: University of Chicago Press, 1970.
 - Lakatos, Imre. *The Methodology of Scientific Research Programmes*. Cambridge: Cambridge University Press, 1978.
 - Feyerabend, Paul. *Against Method*. London: Verso, 1975.
 - Cartwright, Nancy. *How the Laws of Physics Lie*. Oxford: Clarendon Press, 1983.
 - Ladyman, James, and Don Ross. *Every Thing Must Go: Metaphysics Naturalized*. Oxford: Oxford University Press, 2007.
 - Reichenbach, Hans. *The Philosophy of Space and Time*. New York: Dover, 1958.
 - Carnap, Rudolf. *Philosophical Foundations of Physics*. New York: Basic Books, 1966.
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IV. Modern Physics Textbooks

Classical Mechanics:

- Halliday, David, Robert Resnick, and Jearl Walker. *Fundamentals of Physics*. Hoboken, NJ: Wiley, 2018.

- Kleppner, Daniel, and Robert Kolenkow. *An Introduction to Mechanics*. 2nd ed. Cambridge: Cambridge University Press, 2014.
- Goldstein, Herbert. *Classical Mechanics*. 3rd ed. Boston: Addison-Wesley, 2002.
- Landau, L. D., and E. M. Lifshitz. *Mechanics*. 3rd ed. Oxford: Pergamon Press, 1976.
- Feynman, Richard P. *The Feynman Lectures on Physics*, Vol. I. Redwood City, CA: Addison-Wesley, 1963.

Electromagnetism:

- Griffiths, David J. *Introduction to Electrodynamics*. 4th ed. Cambridge: Cambridge University Press, 2017.
- Jackson, John D. *Classical Electrodynamics*. 3rd ed. Hoboken, NJ: Wiley, 1999.

Thermodynamics & Statistical Mechanics:

- Schroeder, Daniel V. *An Introduction to Thermal Physics*. San Francisco: Addison-Wesley, 2000.
- Reif, Frederick. *Fundamentals of Statistical and Thermal Physics*. New York: McGraw-Hill, 1965.

Relativity & Cosmology:

- Taylor, Edwin F., and John A. Wheeler. *Spacetime Physics*. 2nd ed. New York: W. H. Freeman, 1992.
- Schutz, Bernard. *A First Course in General Relativity*. 3rd ed. Cambridge: Cambridge University Press, 2022.
- Carroll, Sean. *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge: Cambridge University Press, 2019.
- Ryden, Barbara. *Introduction to Cosmology*. 2nd ed. Cambridge: Cambridge University Press, 2016.

Quantum Mechanics & Field Theory:

- Griffiths, David J. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2018.
 - Sakurai, J. J., and Jim Napolitano. *Modern Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2020.
 - Dirac, Paul A. M. *The Principles of Quantum Mechanics*. Oxford: Clarendon Press, 1958.
 - Peskin, Michael E., and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. Boulder, CO: Westview Press, 1995.
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